Observer-Based Compensation of Additive Periodic Torque Disturbances in Permanent Magnet Motors

Michael Ruderman, *Member, IEEE*, Alex Ruderman, *Member, IEEE*, and Torsten Bertram

***Abstract*—The impact of additive periodic torque disturbances on the controlled motion of permanent magnet motors can be significant. The paper shows how an observer-based drive control can efficiently reject the harmonic torque disturbances providing smooth angular velocity. The proposed control design is based on the state-space torque harmonics representation and Luenberger observer that proved to be adequate. The designed control algorithms are verified using an experimental setup with a permanent magnet synchronous motor with well-detectable torque harmonics. The rejection of additive position periodic torque disturbances is experimentally demonstrated for two first harmonics and that for different angular velocities.**

***Index Terms*—Cogging torque, harmonics, motion control, observer design, periodic disturbances, permanent magnet motor.**

I. Introduction

HIGH-PERFORMANCE motor control applications like semiconductor wafer production, servo writers, and other close tolerance applications require smooth velocities that may be reached by "flat" ripple-free motor torques. For modern drives, inherent motor torque disturbance compensation can be achieved by appropriate control algorithms at virtually no cost. Therefore, from a system perspective, sometimes it does not make much sense to reduce motor torque ripples at the design stage because it may be costly and result in motor torque/power reduction.

Modern microprocessor-based digital controllers allow implementing more sophisticated algorithms to compensate for torque disturbance effects present in servo drive systems. Moreover, the fast real-time communication architectures [1], [2] integrated in electrical drives permit developing the multi-axis motion controls and decentralizing the control tasks according to the application requirements. This way, the complexity of advanced model-based control algorithms, however, may increase depending on the overall plant disturbances and nonlinearities description (e.g., in [3]) detalization extent.

To compensate for periodic motor toque disturbances, different control strategies have been developed. Various observer-based approaches are reported in [4]–[9]. In [4], the motor dynamics and the transformation into the estimated rotor frame are nonlinear (mainly due to trigonometrical functions), and, thus, the observer and observer error dynamics are nonlinear as well. Two kinds of state observers for speed and acceleration estimation are considered in [5]—the Luenberger full-order observer and an adaptive observer. In [6], a load torque observer has been designed and applied in a torque feed-forward loop to compensate for torque ripples in direct drive motors. The state-space representation and estimation of ripple force for linear hybrid step motors is discussed in [7]. A disturbance torque estimation using the so-called synchronous reference frame-filtering observer has been shown in [8] for low-resolution position interfaces in permanent magnet synchronous machines. Another instantaneous torque estimation within the $d-q$ frame has been proposed with the corresponding torque ripples compensation in [9]. Recently, an adaptive Luenberger disturbance observer has been proposed in [10] in order to compensate for dynamic disturbances caused by the parameter variations or other unstructured uncertainties. Also, an adaptive interconnected observer in the stationary $\alpha-\beta$ reference frame has been recently proposed in [11] for the sensorless control of a synchronous motor.

An explicit modeling and identification of flux linkage harmonics and slot harmonic torque has been proposed in [12] with a compensation scheme applied to high-precision permanent magnet motor drives. Later, in [13], a high-bandwidth deadbeat current control has been used by the authors to compensate for motor toque ripples. The direct torque control with an explicit reduction of the torque ripples is described in [14]–[17]. The method proposed in [14] introduces a fuzzy adaptive modulation technique which may be applied to any induction motor. In [15], the direct torque control is applied to brushless dc drives and the reduced torque ripple and faster dynamic response are shown in comparison with a conventional current control. The work in [16] presented a prediction scheme to diminish both the torque and flux ripples in a direct torque control of induction motor drives. A modified direct torque control for permanent-magnet synchronous machines which uses voltage vectors with variable amplitude and angle enabling the torque and flux-ripple reduction is proposed in [17]. The special switching sequences for space vector-based pulsewidth modulation (PWM) are used in [18] in order to reduce the current and hence torque ripples. Furthermore, compensation strategies which imply the Fourier analysis of torque ripples are, e.g., in [19] and [20].

While position-dependent torque harmonics order and count are typically supposed known *a priori*, in [21], it was shown that also one unknown disturbance frequency can be identified and compensated using an internal state-space harmonic representation and an appropriate Lyapunov-based adaptation law. For

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multiple harmonics it is problematic to identify simultaneously the frequency, amplitude, and phase, though a theory for compensation of single biased harmonics with all three unknown parameters is provided, e.g., in [22].

This paper is focused on compensation of multiple additive position periodic torque disturbances similar to pure cogging and "quasi-cogging" due to magnetic unbalance pull discussed below. At the same time, nonsinusoidal back EMF (electromotive force) distribution can result in multiplicative position periodic torque disturbances that means a motor current multiplication. Compensation of such multiplicative disturbances is beyond the scope of this paper. The proposed observer-based compensation is inspired by approach [21] for position periodic additive disturbances compensation. This paper proposes an implementation of the state-space model of multiple additive position periodic torque disturbances and a suitable Luenberger observer representation including observer gains. The outline of the paper is the following. In Section II, the backgrounds of the additive position periodic torque disturbances including cogging in permanent magnet synchronous motors (PMSMs) are given. The state observation of a PMSM drive is addressed in Section III that introduces the corresponding model and analyses of suggested Luenberger state observer. The experimental PMSM drive taken for evaluation is described in Section IV. The control structure and design of the underlying robust feedback controller are given in Section V. The main experimental observation and compensation results are reported in Section VI. At the end, the concluding remarks are summarized in Section VII.

II. Additive Position Periodic Toque Disturbances in Permanent Magnet Motors

Here, we present a concentrated form a summary of additive harmonic torques in permanent magnet surface mount motors due to teeth-slot stator structure giving references to the original papers. This covers the classic cogging torques that are magnetic reluctance torques by nature and those due to unbalanced magnetic pull. It is suggested that the compensation method proposed in the paper will work equally for all other observable additive torque harmonics.

Cogging torque is a parasitic reluctance torque in permanent magnet brushless motors. Most industrial brushless motors are three-phase with surface mounted permanent magnets in rotor. For a three-phase motor, a slot (teeth) count is a multiple of three. Motors with overlapping concentrated windings have integral slot-to-pole ratio (equal 3) like four poles and 12 slots, six poles and 18 slots, eight poles and 24 slots. Cogging in such motors is high and most straightforward and easy to understand—they have six cogging periods per electrical revolution. For motors with nonoverlapping concentrated windings that have a fractional slot-pole ratio, cogging period analysis must be based on finding zero cogging points due to symmetry considerations. It may be shown that a count of cogging periods on one mechanical revolution is a minimal multiple of poles and slots count (LCM—Least Common Multiple) [23]. For example, a four-pole motor with six stator slots $(\mathrm{LCM} = 12)$, six-pole with nine slots $(\mathrm{LCM} = 18)$, eight-pole with 12 slots

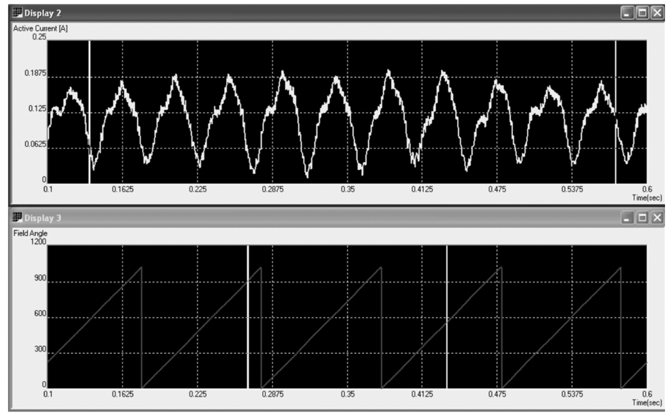


Fig. 1. Torque disturbance due to unbalanced magnetic pull of a nine-slot eight-pole motor on five electrical periods.

$(\mathrm{LCM} = 24)$, ten-pole with 15 slots $(\mathrm{LCM} = 30)$ all have 3/2 slot-pole ratio and six (LCM divided by the pole pairs count P/2) cogging periods on one electrical revolution. In such motors cogging torque is still sufficiently high.

In motors with slot-to-pole ratio (S/P) equal to 6/8, 9/8, 9/10, etc., the LCM is relatively high and the cogging torque is relatively low. These motor designs have a disadvantage of an unbalanced magnetic pull due to the diametrically asymmetric disposition of the stator slots and coils [24]. This way, while the classic cogging is low, torque disturbance due to magnetic pull ("quasi-cogging") is still a problem. In general, unbalanced magnetic pull may have an adverse effect of reduced bearings lifetime, increased vibration, and acoustic noise. For example, a motor that has $\mathrm{P} = 8$ and $\mathrm{S} = 9$ according to the LCM theory must show 72 cogging periods on a mechanical revolution (18 cogging periods on electric revolution). However, its measured torque disturbance looks like a combination of two harmonics—high frequency with small amplitude and dominating low-position harmonic having nine periods on a mechanical revolution, or $9/4 = 2.25$ periods on electrical revolution (see in Fig. 1). This dominating low-frequency torque disturbance is due to unbalanced magnetic pull. This way, to have a magnetically balanced three-phase machine, the practical slot-and-pole combinations must be selected from the following row—6/4, 6/8, 12/10, 12/14, 18/16, 18/20, etc. In addition to appropriate selection of slot and pole counts, motor design techniques aimed at cogging reduction include skewing rotor magnets or stator laminations. Here, the idea is to have local cogging torques averaged out over motor active length [23]. Skewing of rotor magnets or stator laminations may increase production cost and motor torque/power output.

All said above applies to iron-core brushless motors with tooth-slot stator structure. Cogging in its classic sense is eliminated in slotless iron-core motors and in iron-less motors. However, it comes at the expense of reduced torque/power density. However, these types of motors may also show relatively low position periodic torque disturbances of different physical nature. For instance, low-order harmonics appear due to eccentricities of the mounted motor armature.

III. PMSM Observation

A. State-Space Model

Consider a most common dynamic model

$$J\ddot{q}(t) + \vartheta\dot{q}(t) + D\left(q(t)\right) = K_m \left(1 - \mathrm{e}^{-\frac{t}{T_c}}\right) i_r(t) \quad (1)$$

of an electric machine with regulated active current. The inertia and damping factor are denoted by J and ϑ correspondingly. $D(q)$ represents the PMSM position dependent torque disturbance. On the right-hand side of (1), the first-order time delay element with the time constant T_c is assumed. When $T_c \ll \min\{T_i\}$, where T_i constitutes all of the residual time constants of the PMSM drive, the first-order time delay behavior of the current control loop can be neglected. Thus, the right-hand side of (1) becomes the simple excitation with no dynamics, realized by the reference current i_r which is gained by the motor constant K_m. Although a PMSM drive is considered it can be noted that any loaded permanent magnet motor with a fast current regulation and no significant eigenbehavior of the electrical part can be approximated by (1). The dynamic variables q and $\dot{q}$ constitute the observable system states and are the angular position and angular velocity of the rotor shaft, respectively.

Even if individual torque harmonics are not of a straight sinewave shape, their additive coaction can be approximated by a Fourier series expansion

$$D(q) = d_0 + \sum_{n \subseteq \mathbb{N}_0} A_n \sin(nq + \varphi_n) = \sum_{i=0}^{k} d_i(q). \quad (2)$$

Here, A_n is the amplitude and φ_n is the phase of the nth torque harmonic. A total of k harmonics are considered and assumed to be of the known frequencies—multiple of the fundamental rotary one. Note that d_0 constitutes the dc bias which is assumed to be constant during a unidirectional motion. In most cases, the bias value can be attributed to the Coulomb friction and described by $d_0 = A_0 \mathrm{sgn}(\dot{q})$.

Keeping in mind that the system dynamics are given as a function of time and not angular position, the single torque harmonic can be written as

$$d(t) = A \sin\left(n\dot{q}(t)t + \varphi\right). \quad (3)$$

It is evident that the harmonic oscillation equation (3) represents a general homogeneous time-domain solution of an undamped second-order linear system

$$\ddot{d} + n^2 \dot{q}^2 d = 0 \quad (4)$$

and A and φ constitute the initial values of a free oscillation.

Introducing the state variables $x_1 = d$ and $x_2 = \dot{d}$, one obtains the state-space model

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -n^2 \dot{q}^2 x_1. \end{aligned} \quad (5)$$

Assuming the overall state vector $\dot{\mathbf{x}} = (d_1, \dot{d}_1, \ldots, d_k, \dot{d}_k, \dot{q})^T$, the state-space model of the PMSM drive can be noted as

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}i_r + \mathbf{H}\mathrm{sgn}(\dot{q}) \quad (6)$$

which is equivalent to (1) without time delay element on the right-hand side. $\mathbf{A}$ is the $(2k+1) \times (2k+1)$ system matrix (provided later in Section III-B), and $\mathbf{B}$ and $\mathbf{H}$ are the $(2k+1) \times 1$ coupling vectors, one of the input value and one of the system nonlinearities, respectively. Since both the angular velocity and the harmonic torque disturbances constitute the output values of interest, the output equation can be written as

$$\begin{pmatrix} D \\ \dot{q} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} A_0 \\ 0 \end{pmatrix} \mathrm{sgn}(\dot{q}). \quad (7)$$

B. Luenberger State Observer

The well-known Luenberger observer [25] is widely used for estimating the dynamic states of time-invariant linear systems when those are not fully measurable. Using the observer gain $\mathbf{L}$, the state estimate $\hat{\mathbf{x}}$ is given by

$$\dot{\hat{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}y \quad (8)$$

for a single-input single-output (SISO) system. The vector $\mathbf{C}$ transforms the state vector into the measurable output value y which is used to determine the observation error. The input value u excites the observer dynamics equally as the system to be observed. The observer gain weights the observation error $\mathbf{e}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t)$, thus affecting the overall convergence of the state estimate.

The eigenbehavior of the observation error is

$$\dot{\mathbf{e}}(t) = (\mathbf{A} - \mathbf{L}\mathbf{C})\mathbf{e}(t) \quad (9)$$

with an initial value $\mathbf{e}(0) = \mathbf{x}_0 - \hat{\mathbf{x}}_0$. Furthermore, it is valid that

$$\lim_{t \to \infty} \|\mathbf{e}(t)\| = 0 \quad \forall \quad \{\mathbf{x}_0, \hat{\mathbf{x}}_0\} \quad (10)$$

under the single condition that all poles of $\mathbf{A} - \mathbf{L}\mathbf{C}$ matrix are in the negative half-plane. From the applications point of view, the observer gain has to be selected so as to shift the poles of observer possibly far on the left in comparison to those of the system. Otherwise, the state estimate will be always lagged behind the system dynamics during nonsteady-state operation modes. From the other hand, the observer gain has to be bounded with respect to the measurement noise, because of the last summand on the right-hand side of (8).

In terms of observing the linear part of (6), the system matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & \ldots & 0 & 0 & 0 \\ -(n_1\dot{q})^2 & 0 & \ldots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & 1 & 0 \\ 0 & 0 & \ldots & -(n_k\dot{q})^2 & 0 & 0 \\ -1/J & 0 & \ldots & -1/J & 0 & -\vartheta/J \end{pmatrix} \quad (11)$$

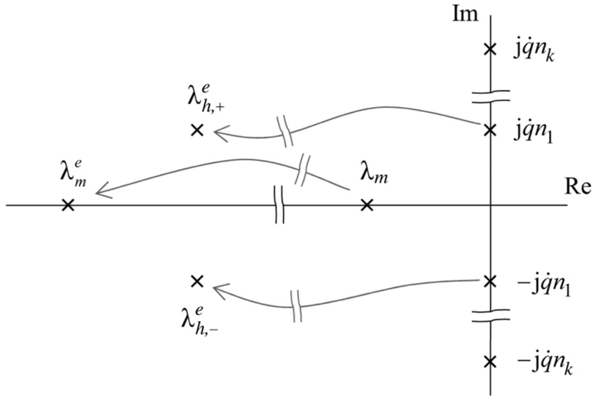


Fig. 2. Poles shifting during observer design. The system poles λ_m and $\pm j\dot{q}n_1$ are transferred into the observation error poles λ^e_m and $\lambda^e_{h,+/-}$.

and the output vector

$$\mathbf{C} = (0, 0, \dots, 0, 0, 1) \tag{12}$$

have to fulfill the Kalman's criterion of system observability

$$\operatorname{rank}(\mathbf{C}, \mathbf{CA}, \mathbf{CA}^2, \dots, \mathbf{CA}^{2k})^T = 2k + 1. \tag{13}$$

For all angular velocities $\dot{q} \neq 0$, it proves to be valid under the single condition $n_i \neq n_j \forall i \neq j$. From the application point of view, it means that closer two harmonics n_i and n_j are to each other, more perturbed is their observation and separation. In an extreme case $n_i = n_j$, the system observation becomes degraded concerning the n_ith and n_jth harmonics.

C. Determining Observer Gain

In order to determine an appropriate observer gain, let us first analyze the eigenbehavior of the linear subsystem given by (11). Apart from the real pole $\lambda_m = -\vartheta/J$, which determines the mechanical time constant of the PMSM drive, the conjugate-complex poles

$$\lambda_h \in \{0 \pm \mathrm{j}\dot{q}n_i\} \tag{14}$$

are related to the torque harmonics $i = [1 \dots k]$. Owing to the observer gain to be determined the modeled system poles are transferred into the observation error poles denoted by λ^e as exemplary, shown in Fig. 2. The real pole λ_m is shifted to the left providing the observation of the angular velocity. A well-directed pole placement [26] can transfer the conjugate-complex poles of torque harmonics (e.g. $\pm \mathrm{j}\dot{q}n_1$) either into the also conjugate-complex poles ($\lambda^e_{h,+/-}$) or into their real counterparts.

As well known, the design of an observer gain constitutes a tradeoff between a preferable fast convergence of the observation error and a measurement noise which is unavoidable when capturing y. The latter one is propagated through the observer gain according to (8).

Let us derive several conditions for determining the observer gain according to the above analysis. First, bearing in mind that each torque harmonic is a multiple of the fundamental rotary frequency, it can be assumed that the estimate of the angular velocity have to converge must faster than that of the torque harmonics. Thus, it is assumed that

$$|\lambda^e_m| \gg \left\|\lambda^e_{h,i}\right\| \quad \forall \quad i = 1 \dots k. \tag{15}$$

Further, assuming that each torque harmonic is proportional to the instantaneous angular velocity, it is worth requiring

$$\left\|\lambda^e_{h,i}\right\| \gg |\dot{q}|n_i \quad \forall \quad i = 1 \dots k. \tag{16}$$

In this regard, the observer gain can be considered as a general function

$$\mathbf{L} = f(\dot{q}, \mathbf{n}) \tag{17}$$

of the instantaneous angular velocity and particular torque harmonics. Note that the selection of an appropriate mapping f depends not only on the plant eigenbehavior (11), but equally on the level of process and measurement noise. In the following, no analytical computation of an optimal f strategy is provided. Instead, the observer gain law is derived based on the introduced conditions (15)–(17) and using the numerical simulation of the modeled PMSM drive with control loop.

IV. Experimental System

A. PMSM Drive

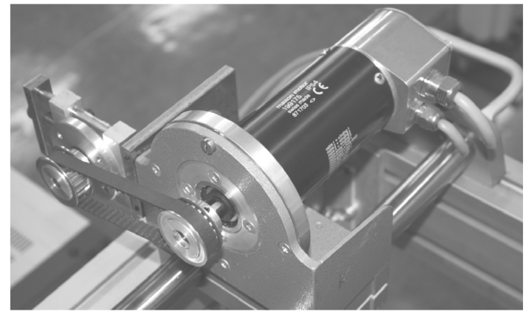


Fig. 3. Experimental setup of PMSM drive with additional pulley load.

The PMSM drive under evaluation is the low-power (400 W) high-dynamic Maxon motor with the number of phases 3 and number of poles 2. Despite the relative trivial phase–pole ratio the overall PMSM drive exhibits several torque harmonics (shown further in Section VI-A) whose exact source analysis is beyond the scope of this work. The PMSM with idling speed 3100 rpm is operated through a PWM electronic unit with an embedded current control. Both the angular position and velocity are captured by the 15 bit resolver mounted directly on the motor shaft. In order to increase the overall damping of the motor idle running, a second pulley is connected to the pulley on the motor shaft through a tension belt. The pulley set provides an additional inertia and damping load, the latter one mainly due to the normally loaded ball-bearings. Due to a relative low inertia of the second pulley (in comparison to the inertia of PMSM) and a relative high stiffness of the stretched belt, no significant elasticities appear in the dynamic system response as approved by the first-order linear approximation provided in Section V-B. The experimental setup of PMSM drive is depicted in Fig. 3.

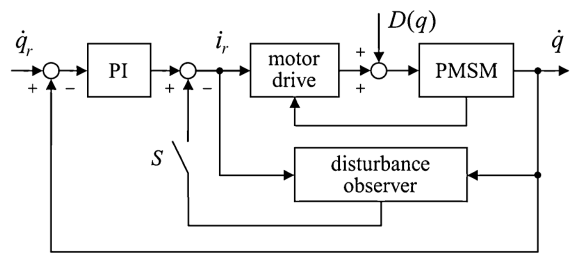


Fig. 4. Block diagram of the velocity control with compensation of additive position periodic torque disturbances.

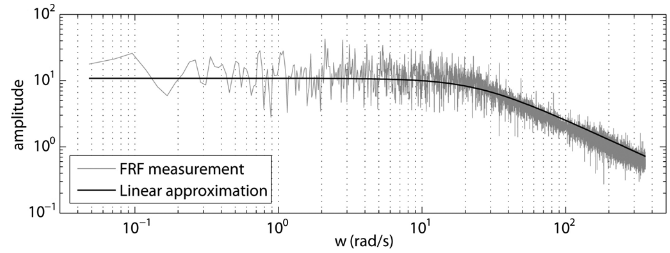


Fig. 5. Measured and linear approximated frequency response function.

B. Digital Drive Controller

Modern industrial digital drives offer several advanced features. The today's industry standard is to support any feedback sensor from incremental and analog sin/cos encoders to absolute serial ones (EnDAT 2.2, Hiperface, BiSS/SSI and other). Excellent servo performance can be achieved due to high control loops bandwidths owing to very fast control loop sampling times—as low as 40 μs for current loop and 80 μs for position and velocity loops. Advanced position control modes include point-to-point, position and position–velocity tables, ECAM follower, dual loop (two position sensors—on motor and on load), and fast event capturing inputs.

Available control infrastructure that includes advanced high-order linear (e.g., low pass, notch, or general biquad) and nonlinear control filters allows to reliably handle different load types and compensate for various disturbances. Fast and easy-to-use auto-tuning procedures allow automatic setting up of optimal control loop gains at the commissioning stage. Modern drives support different communication protocols, the most common of which are EtherCAT, CANopen, USB, and RS-232. Current loop gain scheduling allows for efficient rejection of disturbances due to dc bus voltage variation. Velocity and position loops gain scheduling for boosting the respective loops performance in the presence of mechanical coupling and minimize speed and position tracking errors.

The digital servo drive used in this work is *Solo Whistle* from Elmo Motion Control Ltd. The available CANopen interface allows to connect the digital servo drive to an external laboratory controller. The latter is realized on the basis of xPC Target real-time platform from MathWorks Inc., using the CAN-AC2-PCI adapter from Softing AG company. The automatically tuned high-gain current control loop is running inside of the digital servo drive with an 11-kHz rate. An extensive *SimplIQ* software command interface enables to operate the servo drive in the current control mode and thus to apply an extended external velocity controller. The external controller implements the compensation scheme shown in Fig. 4. All of the control algorithms are realized in the *Simulink* paradigm and directly compiled into the real-time xPC target code. The fixed-step *ode3* solver is used, and the discrete-time forward Euler integrators are taken for the control implementation. All of the signals of the external controller are sampled with a 2-kHz rate and running within the real-time xPC Target environment on the applied external controller with CANopen interface.

V. Control Design

A. Control Structure

The compensation of additive position periodic torque disturbances extends a standard feedback control of the PMSM drive which can operate either in the velocity or in the positioning mode. With respect to cascade control structures, which constitute a most common practical case, a velocity control as schematically shown in Fig. 4 is considered here. A linear PI controller serves as a standard reference regulation of $\dot{q}_r$. Note that an explicit consideration of the inner current control loop is out of scope as established before in Section III-A. The state observer based on (8) provides the compensation value which can be directly injected to the control value using the digital switch S. Note that the constant bias d_0 can be equally included when computing the compensation signal.

In consideration of the system dynamics equation (1), the total biased torque harmonics captured by $D(q)$ act as an input disturbance, whose analytical form is assumed to be known. On the contrary, the parameters and that the amplitude and phase as well as the input q are not available during the compensation. Note that the measured angular position can be however required during a primary system analysis, in order to detect the significant harmonic torque disturbances.

B. Robust Feedback Control

The linear PI reference control is designed using the estimated frequency response function (FRF) of the PMSM drive. The FRF is obtained by applying a random binary signal (RBS) ± 0.5 A with the bandwidth 0.01–20 Hz. For computing the FRF at equidistant frequency points, the well-known correlation H1 method (see, e.g., [27] for details) is used, whereas the data is averaged over ten equivalent measurements, each one 120 s in duration. Afterwards, the assumed first-order linear approximation of the plant is fitted in a least-squares sense within the frequency domain. In Fig. 5, the FRF of the identified linear plant approximation is shown versus the measurement. Note that, for identification purposes, the amplitude response only is used, due to a large discrepancy detected in the measured phase response. The latter one can be attributed to the nonlinear friction and dead times whose impact on the phase response becomes more significant at higher frequencies (see [28] for the impact of nonlinear friction).

The resulting transfer function of the open control loop is

$$\mathrm{PI}(s) \cdot \mathrm{H}(s) = \frac{K_r(T_r s + 1)}{T_r s} \cdot \frac{10.81}{0.042 s + 1} \tag{18}$$

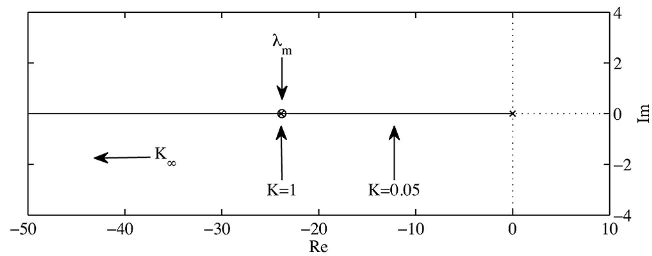


Fig. 6. Root locus of the open control loop.

where the gain K_r and the integral time constant T_r constitute the control parameters to be adjusted. The identified linear approximation of the PMSM drive is denoted by $\mathrm{H}(s)$. For a possibly fast rising time of the closed control loop, the parameter value $T_r = 0.042$ is chosen. Note that, when determining the control parameters, the actuator boundaries have to be respected, in order to ensure the intended control behavior without an unconsidered lag due to the integral windup. The present actuator boundaries are set to ± 6 A according to the rated current.

The robustness of the designed PI control loop is illustrated by means of the root locus diagram depicted in Fig. 6. The total gain $\mathrm{K} = |\mathrm{H}(0)| K_r$ of the open control loop can be seen as a measure of the plant uncertainties and particularly of the damping ϑ. Note also that the inertia J contributes to the denominator of the drive transfer function $\mathrm{H}(s)$, but can solely shift the stable pole λ_m. When J decreases, the pole is shifted to the left, and when J increases the pole is shifted to the right having the limiting value at zero when $J \to \infty$. However, seeing that the overall system damping, which contributes also to the numerator of $\mathrm{H}(s)$, underlies the most uncertainties due to the friction the variations of K appear to be of the most interest. Examining the K values labeled exemplarily in Fig. 6, it can be noted that a relative wide gain variation results in an acceptable pole of the closed control loop. The designed robust control remains conditionally fast, and this is without any transient overshoot behavior.

VI. Compensation of Additive Position Periodic Torque Disturbances

A. Harmonics Detection

In order to detect the most significant harmonic torque disturbances and thus to determine the analytical form of $D(q)$, the bidirectional position controlling experiments are performed. Using a low-gain P-control, a flat reference position slope is tracked in both directions, so that a low-velocity motion profile with a nearly zero acceleration is realized for multiple turns of the PMSM shaft. Assuming that, under these quasi-static conditions, the fed input torque is equal to the disturbance torque value, the latter one is determined from the measured motor current. Note that the integral of the disturbances which are related to the torque harmonics must be zero over one mechanical revolution. Thus the computed average value over an integer number of turns provides the measure of dc bias. The bidirectional experiments allow to determine the bias value in a more accurate way.

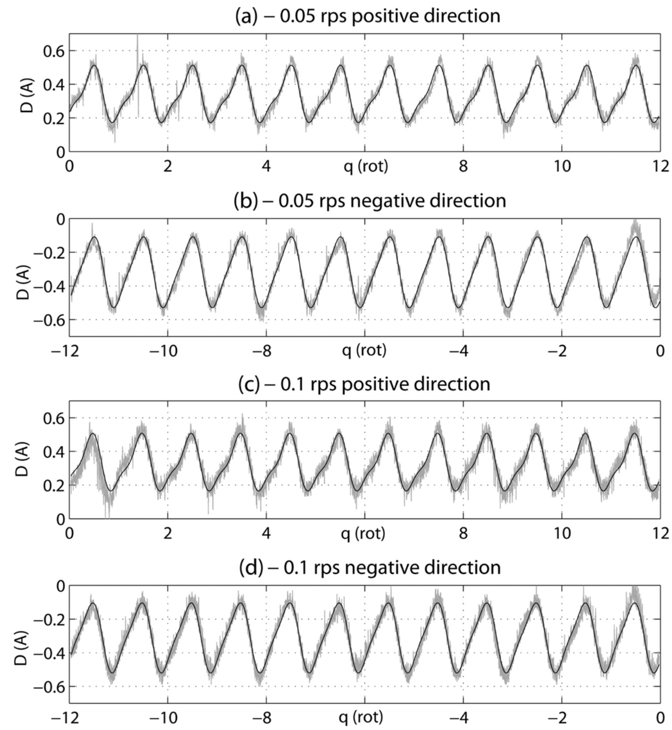


Fig. 7. Measured disturbances over the driven angular position; measurements are depicted by the grey line; the black line represents the model fit.

TABLE I
Disturbing Torque Parameters

Motion	A_0 (A)	A_1 (A)	φ_1 (rot)	A_2 (A)	φ_2 (rot)
0.05 rps pos.	0.3330	0.1487	-0.2007	0.0476	0.1476
0.05 rps neg.	-0.3170	0.1970	-0.2043	0.0399	0.0832
0.1 rps pos.	0.3248	0.1507	-0.1740	0.0470	0.2116
0.1 rps neg.	-0.3084	0.1933	-0.1706	0.0414	0.1417

The controlled motion with two low velocities of 0.05 and 0.1 rps is executed in both directions. The captured disturbance values recorded over the driven turns of the shaft are shown in Fig. 7. The first and second harmonics are detected so that the disturbing torque model eqaution (2) with overall five parameters (including the dc bias) can be directly fitted in a least-squares sense for evaluating each measurement separately. The fitted model response is also shown in Fig. 7. The determined parameters listed in Table I disclose the certain variations, also for the same motion direction. Apparently, the amplitude and phase of each torque harmonic depend not only on the system states, i.e., current and angular velocity, but also on the initial conditions. This fact argues for the use of an observer rather than of a static mapping which must be determined in an offline manner.

In order to evaluate the impact of the torque harmonics on the output angular velocity, the steady-state motion experiments are performed using the linear feedback control described in Section V-B. The experimental data from the drives with a constant reference velocity are taken for determining the amplitude spectrum using the fast Fourier transformation. The amplitude spectra obtained for $\dot{q}_r = \{1, 5, 10\}$ rps are depicted in Fig. 8.

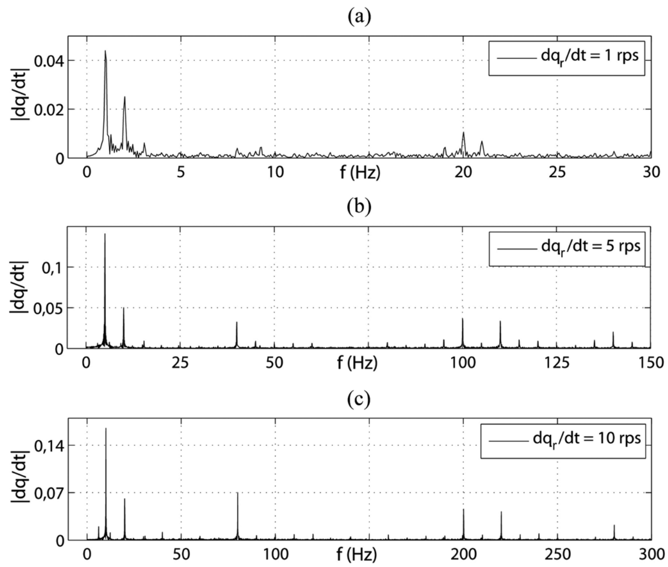


Fig. 8. Measured amplitude spectra of controlled angular velocity.

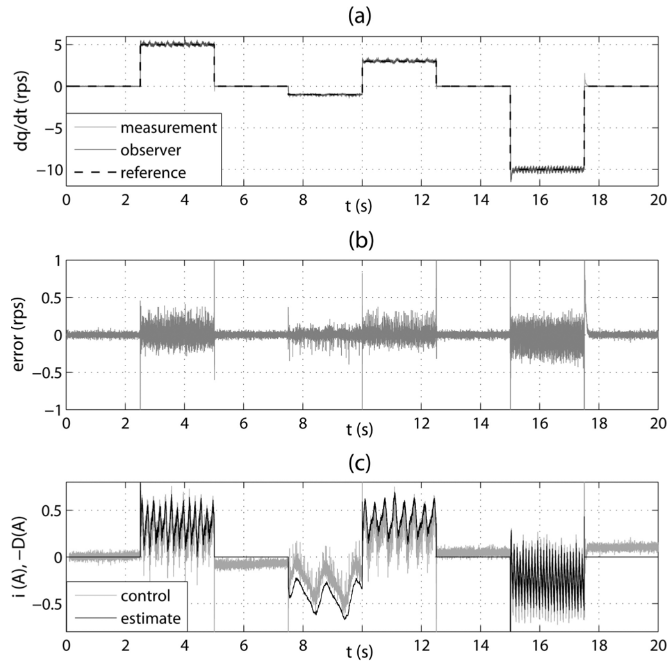


Fig. 9. Measured observer response within closed control loop: (a) velocity, (b) velocity observation error, and (c) control current versus estimated harmonics.

It can be seen that in case of all the steady-state velocities the first and second harmonic are the most significant ones. Furthermore, it can be noted that for higher angular velocities the 8th-, 20th-, and 22th-harmonic gain in impact within the amplitude spectra. However, these ones are not considered here, since the higher order harmonics are not clearly separable from the overall process and measurement noise at higher angular velocities, particularly taking into account the given sampling frame.

B. Observation Results

In order to evaluate the proposed observer in the practice-related sense, the PMSM drive is actuated in the closed-loop manner using the designed PI velocity control. Thus, the observer can be evaluated concerning its robustness against the noisy process signals, so as to be integrated in the control loop according to the block diagram provided in Fig. 4.

The applied observer gain is suggested as

$$L_{2i-1,2i} = (100n_i, 50n_i)^T|\dot{q}| \tag{19}$$

and

$$L_{2k+1} = 100|\dot{q}| + 2000 \tag{20}$$

with respect to the analysis carried out before in Section III-C. Recall that the convergence of the velocity estimate has to be significantly faster than this one of the harmonic states. Furthermore, the gains of two states which represent one and the same harmonic have to be sufficiently differing so as to ensure the estimate of both, unknown amplitude and phase.

In order to ensure the non-divergence of observer at zero velocity (see Section III-B), for which the exciting input can however deviate from zero, the threshold velocity is set to $|\dot{q}_z| =$ 0.1 rps. Below this value, which is of the order of measurement noise, all the integrators of the state-space model are reset to zero. It turns out to be not disadvantageous at all, since a fast convergence of observed states is given, even when the motion direction changes abruptly.

The observer response evaluated experimentally is shown in Fig. 9. The designed PI control tracks the reference trajectory which consists of the sequence of different steps with and without return to zero as shown in Fig. 9(a). The observer receives the measured control signal and output velocity and estimates the overall state dynamics according to (8). Note that the bounded velocity observation error depicted in Fig. 9(b) is not much higher than the overall measurement noise. The control effort is compared with the estimated harmonics in Fig. 9(c). The latter are taken with negative sign and include the dc bias. It is evident that at steady-state the control effort has to coincide with the overall estimated torque disturbances. Inspecting the torque ripples in Fig. 9(c), it proves to be the case for all driving velocities between 1 and 10 rps. A slight deviation occurs as a bias value mainly is attributed to uncertainties in the frictional behavior. Note that, every time the motion direction changes (and hence the observer is reset), the fast observer convergence is reached again without spending much time of the controlled process.

C. Compensation Results

The compensation of harmonic torque disturbances is first evaluated on three different velocity steps 10, 5, and 1 rps. The designed PI control serves as a basic one and the torque harmonics observer is switched-on $(S = 1)$ at time $t =$ 10 s. The compensation results are shown in Fig. 10. The left diagrams (a), (c), and (e) visualize the steady-state close to the observer switch-on time. On the right-hand side in (b), (d), and (f), the amplitude spectra of steady-state velocities are shown for uncompensated and compensated harmonics. In order to make the considered harmonics better separable from the overall noisy measurement the depicted velocity time series (but not the frequency response) are low-pass filtered with the cutoff frequency

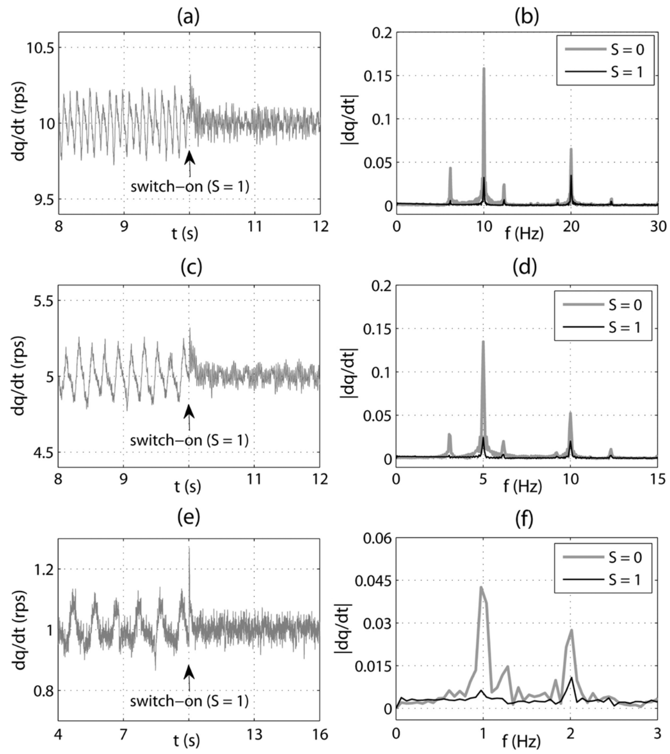


Fig. 10. Compensation results for 10-, 5-, and 1-rps velocities: (a), (c), (e) time series close to observer switch-on and (b), (d), (f) amplitude spectra.

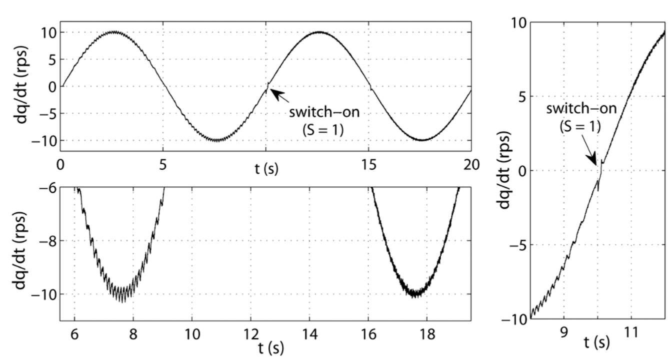


Fig. 11. Compensation results for sinusoidal velocity profile.

100 Hz. It can be recognized in both the time and frequency domain that the considered harmonics are substantially reduced and that for all three driven velocities.

Further, in order to evaluate the rejection of position periodic torque disturbances at continuously changing velocities the reference signal $\dot{q}_r(t) = 10\sin(0.2\pi t)$ is applied. The observer based compensation is switched-on $(S = 1)$ after one full period at time $t =$ 10 s. The overall measured velocity response is shown in Fig. 11 together with close-ups depicted for better recognizing the disturbing ripples. Both the increasing slope as well as the sinus peaks disclose a definite rejection of disturbance harmonics.

VII. Conclusion

In this paper, the additive position periodic torque disturbances in permanent magnet motors are addressed and it is shown how the low-order harmonics can be practically compensated by an observer-based drive control. The proposed simple control algorithm can help to avoid torque disturbances reduction by motor design which may be costly and not easy from a production perspective. Based on Fourier series expansion of additive position periodic torque disturbances, an internal state-space model is derived. The torque harmonic states are proved to be observable at known frequencies. This allows formulating the appropriate conditions for designing the observer gains.

For the selected PMSM drive experimental setup, the two first and most significant torque harmonics are detected and afterwards rejected using the proposed compensation scheme. The experimental evaluation has shown a similar compensation efficiency for different angular velocities (between 1 and 10 rps). Besides that, a stable pass through zero velocity (at which the system observability is degraded) is realized, and thereby a fast convergence of observer is shown. In principle, this allows applying the compensation scheme also for dynamic velocity follower control with frequent motion reversals. It is suggested and proved theoretically that a compensation for additive position periodic torque disturbances of higher order harmonics is feasible in similar way.

References

[1] F. Benzi, G. Buja, and M. Felser, "Communication architectures for electrical drives," *IEEE Trans. Ind. Inf.*, vol. 1, no. 1, pp. 47–53, Feb. 2005.

[2] K. Kim, M. Sung, and H. Jin, "Design and implementation of a delay-guaranteed motor drive for precision motion control," *IEEE Trans. Ind. Inf.*, vol. 8, no. 2, pp. 351–365, May 2012.

[3] M. Ruderman, F. Hoffmann, and T. Bertram, "Modeling and identification of elastic robot joints with hysteresis and backlash," *IEEE Trans. Ind. Electron.*, vol. 56, no. 10, pp. 3840–3847, Oct. 2009.

[4] L. Jones and J. Lang, "A state observer for the permanent-magnet synchronous motor," *IEEE Trans. Ind. Electron.*, vol. 36, no. 3, pp. 374–382, Aug. 1989.

[5] P. Nandam and P. Sen, "A comparative study of a Luenberger observer and adaptive observer-based variable structure speed control system using a self-controlled synchronous motor," *IEEE Trans. Ind. Electron.*, vol. 37, no. 2, pp. 127–132, Apr. 1990.

[6] N. Matsui, T. Makino, and H. Satoh, "Autocompensation of torque ripple of direct drive motor by torque observer," *IEEE Trans. Ind. Appl.*, vol. 29, no. 1, pp. 187–194, Jan/Feb 1993.

[7] T.-S. Hwang and J.-K. Seok, "Observer-based ripple force compensation for linear hybrid stepping motor drives," *IEEE Trans. Ind. Electron*, vol. 54, no. 5, pp. 2417–2424, May 2007.

[8] T. Tesch and R. Lorenz, "Disturbance torque and motion state estimation with low-resolution position interfaces using heterodyning observers," *IEEE Trans. Ind. Appl.*, vol. 44, no. 1, pp. 124–134, Jan.–Feb. 2008.

[9] N. Nakao and K. Akatsu, "Torque ripple compensation based on instantaneous torque estimation in permanent magnet synchronous motors," in *Proc. IEEE Int. Electric Machines Drives Conf.*, May 2011, pp. 989–994.

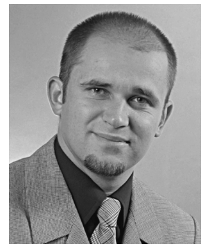
[10] C. Xia, M. Wang, Z. Song, and T. Liu, "Robust model predictive current control of three-phase voltage source PWM rectifier with online disturbance observation," *IEEE Trans. Ind. Inf.*, vol. 8, no. 3, pp. 459–471, Aug 2012.

[11] M. A. Hamida, J. De Leon, A. Glumineau, and R. Boisliveau, "An adaptive interconnected observer for sensorless control of PM synchronous motors with online parameter identification," *IEEE Trans. Ind. Electron.*, vol. 60, no. 2, pp. 739–748, Feb. 2013.

[12] J. Holtz and L. Springob, "Identification and compensation of torque ripple in high-precision permanent magnet motor drives," *IEEE Trans. Ind. Electron.*, vol. 43, no. 2, pp. 309–320, Apr. 1996.

[13] L. Springob and J. Holtz, "High-bandwidth current control for torqueripple compensation in PM synchronous machines," *IEEE Trans. Ind. Electron.*, vol. 45, no. 5, pp. 713–721, Oct. 1998.

[14] L. Romeral, A. Arias, E. Aldabas, and M. Jayne, "Novel direct torque control (DTC) scheme with fuzzy adaptive torque-ripple reduction," *IEEE Trans. Ind. Electron.*, vol. 50, no. 3, pp. 487–492, Jun. 2003.
[15] Y. Liu, Z. Zhu, and D. Howe, "Direct torque control of brushless DC drives with reduced torque ripple," *IEEE Trans. Ind. Appl.*, vol. 41, no. 2, pp. 599–608, 2005.
[16] J. Beerten, J. Verveckken, and J. Driesen, "Predictive direct torque control for flux and torque ripple reduction," *IEEE Trans. Ind. Electron.*, vol. 57, no. 1, pp. 404–412, Jan. 2010.
[17] Y. Zhang, J. Zhu, W. Xu, and Y. Guo, "A simple method to reduce torque ripple in direct torque-controlled permanent-magnet synchronous motor by using vectors with variable amplitude and angle," *IEEE Trans. Ind. Electron.*, vol. 58, no. 7, pp. 2848–2859, Jul. 2011.
[18] G. Narayanan, V. Ranganathan, D. Zhao, H. Krishnamurthy, and R. Ayyanar, "Space vector based hybrid PWM techniques for reduced current ripple," *IEEE Trans. Ind. Electron.*, vol. 55, no. 4, pp. 1614–1627, Apr. 2008.
[19] M. Ruderman, F. Hoffmann, J. Krettek, and T. Bertram, "Detection and self-tuning compensation of periodic disturbances by the control of DC motor," in *Proc. 13th Power Electron. Motion Control Conf.*, 2008, pp. 1215–1220.
[20] Y. Mandel and G. Weiss, "Reduction of torque ripple in brushless DC motor drives," in *Proc. 6th IFAC Symp. Robust Control Design*, 2009, pp. 84–89.
[21] C. Canudas de Wit and L. Praly, "Adaptive eccentricity compensation," *IEEE Trans. Control Syst. Technol.*, vol. 8, no. 5, pp. 757–766, May 2000.
[22] A. A. Bobtsov, "Output control algorithm with the compensation of biased harmonic disturbances," *Automation and Remote Control*, vol. 69, no. 8, pp. 1289–1296, 2008.
[23] Z. Zhu and D. Howe, "Influence of design parameters on cogging torque in permanent magnet machines," *IEEE Trans. Energy Conv.*, vol. 15, no. 4, pp. 407–412, 2000.
[24] J. Hendershot and T. Miller*, Design of Brushless Permanent-Magnet Motors*, ser. Monographs in electrical and electronic engineering. New York: Magna Pysics Publishing and Clarendon Press, 1994.
[25] D. G. Luenberger, "Observing the state of a linear system," *IEEE Trans. Military Electron.*, vol. 8, no. 2, pp. 74–80, 1964.
[26] J. Kautsky, N. K. Nichols, and P. V. Dooren, "Robust pole assignment in linear state feedback," *Int. J. Control*, vol. 41, no. 5, pp. 1129–1155, 1985.
[27] T. P. Dobrowiecki and J. Schoukens, "Practical choices in the FRF measurement in presence of nonlinear distortions," *IEEE Trans. Instrum. Meas.*, vol. 50, no. 1, pp. 2–7, 2001.
[28] M. Ruderman and T. Bertram, "FRF based identification of dynamic friction using two-state friction model with elasto-plasticity," in *Proc. IEEE Int. Conf. Mechatron.*, Istanbul, Turkey, 2011, pp. 230–235.



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